4-1 Mains oxygen failure v.1

Complete failure of wall or pendant oxygen supply. Failure may be theatre-specific, zone-specific or hospital-wide.

START

1. Inform the theatre team of the problem, including theatre coordinator (Box A).
2. Call for help, but be aware failure may be widespread.
3. Switch to cylinder oxygen supply.
4. Check remaining cylinder content (Box B):
   - Ensure at least one further cylinder always available.
5. Prolong duration of cylinder oxygen supply:
   - Use circle system.
   - Use lowest possible fresh gas flow.
   - Use lowest possible FiO_2.
   - Check FiO_2 alarms.
6. Minimise auxiliary gas usage:
   - Some ventilator systems are gas-driven. If so, switching to manual ventilation will prolong oxygen cylinder supply.
7. Evaluate moving to nearby area if oxygen supply there preserved.
8. Continue to monitor and ensure adequacy of inspired oxygen and volatile agent.
9. Workflow:
   - Do not start any new case unless clinical priority absolutely requires it.
   - Expedite conclusion of current case if clinically appropriate.
   - Consider recovering the patient in theatre if recovery also affected (Box C).
10. Disconnect from failed outlets and do not re-connect until safe to do so (Box D).

Box A: ORGANISATIONAL ISSUES

Theatre coordinator or equivalent should:

- Ensure no new case is started unless clinical priority absolutely requires it.
- Ascertain the extent of the failure throughout the hospital.
- Ascertain the reserve supplies of oxygen.
- Evaluate implications for ongoing supply.
- Ensure any other relevant emergency plans are initiated.
- Coordinate delivery of further oxygen cylinders in good time.
- Ensure individual theatres are kept informed.

Box B: OXYGEN CYLINDER: REMAINING CONTENTS

% full (pressure)	CYLINDER SIZE					
	C	D	E	F	G	J
100% (137 bar)	170 L	340 L	680 L	1360 L	3400 L	6800 L
50% (69 bar)	85 L	170 L	340 L	680 L	1700L	3400 L
25% (34 bar)	43 L	85 L	170 L	340 L	850 L	1700L

Box C: RECOVERY

- Recovery area must be appropriately supplied with cylinder oxygen if also affected.
- Identify patients who do not require supplemental oxygen.

Box D: RE-ESTABLISHMENT OF SERVICES

- Disconnect equipment from failed wall/pendant outlets.
- When re-established, output may not initially be 100% oxygen.
- Do not re-use outlet until gas composition and quality satisfactorily confirmed by a competent authority.

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